Write a python code for the following questions.

Bayes' Rule

1. Student Department Problem

In a college:

- 60% of students belong to the Science department.
- 40% belong to the Arts department.
- 20% of Science students participate in a programming competition.
- 10% of Arts students participate in the competition.

2.Email Spam Problem

In an email system, 30% of incoming emails are spam. A particular word appears in 80% of spam emails and in 10% of non-spam emails.

If an email contains this word, calculate the probability that the email is spam using Bayes' Rule.

Binomial Distribution

1. A fair coin is tossed 10 times. Find the probability of getting:
    - ○ exactly 6 heads
    - ○ exactly 4 tails
    - ○ at least 1 head
2. A multiple-choice question has 4 options, with only one correct answer. If a student guesses the answers to 8 questions, find the probability that the student gets exactly 3 correct answers.
3. A factory produces electronic components, and 5% of the components are defective. If 10 components are selected randomly, find the probability that:
    - ○ exactly 2 are defective
    - ○ no component is defective
    - ○ at least 1 component is defective

Uniform Distribution

1. The lifetime of a particular battery is uniformly distributed between 500 and 1,000 hours. Calculate the probability that a randomly selected battery lasts:
    - ◯ less than 600 hours
    - ◯ more than 800 hours
    - ◯ between 600 and 900 hours.
2. The time required to download a file is uniformly distributed between 10 and 30 seconds. Find the probability that the download takes:
    - ◯ less than 15 seconds
    - ◯ more than 25 seconds
    - ◯ between 15 and 25 seconds.
3. A bus arrives at a bus stop at a random time between 8:00 AM and 8:20 AM. Assuming the arrival time is uniformly distributed, find the probability that a passenger arriving randomly will wait:
    - ◯ less than 5 minutes
    - ◯ more than 10 minutes
    - ◯ between 5 and 15 minutes.

Poisson Distribution

1. A call center receives an average of 4 calls per minute. Assuming the number of calls follows a Poisson distribution, find the probability of receiving:
    - ◯ exactly 2 calls in one minute
    - ◯ exactly 5 calls in one minute
    - ◯ no calls in one minute.
2. A hospital emergency department receives an average of 6 patients per hour. Find the probability that in a randomly selected hour:
    - ◯ exactly 4 patients arrive

- ◯ exactly 8 patients arrive
- ◯ no patients arrive.

3. A website receives an average of 10 visitors per minute. Using the Poisson distribution, calculate the probability of receiving:
   - ◯ exactly 8 visitors in one minute
   - ◯ more than 10 visitors in one minute
   - ◯ fewer than 3 visitors in one minute.

Mean, Variance and Standard deviation

1. Two students are applying for a scholarship. Their marks in five tests are:
   - ◯ Student A: 70, 72, 68, 71, 69
   - ◯ Student B: 55, 85, 65, 90, 70

Calculate the mean, variance, standard deviation for both students.

2. A quality-control department measures the diameter of a manufactured component in millimetres:
   10.2, 10.1, 10.3, 10.2, 10.4
   Calculate the mean, variance and standard deviation.